



Nurturing regenerative food systems in a changing climate

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S20.3 - Using Decision Analysis to Support Agricultural Land Use Policy in Peri-Urban Hanoi: Integrating Local Knowledge and Validation

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Abstract: Rapid urbanization and demographic shifts in peri-urban Hanoi are driving a decline in farming interest and an aging rural workforce, leading to increasing land abandonment. This study applied decision analysis to examine how farming households navigate among three land use options: (1) continuing cultivation, (2) renting out land, and/or (3) leaving land fallow.

We developed a probabilistic decision analysis model that integrates causal mapping, expert elicitation, and Monte Carlo simulations to estimate the net present value (NPV) of each decision pathway. The initial model was informed by a review of 15 peer-reviewed studies and subsequently refined through focus groups involving 50 farmers from two districts in Hanoi. Participants received training to recognize and mitigate common biases in data estimation, using culturally appropriate examples. Following the training, participants provided data inputs with 90% confidence intervals.

The model was iteratively refined through community feedback and expert consultations. These consultations provided valuable insights into local interpretations of key risks, including market instability, labor constraints, land use conflicts, and ambiguous land use policies. This input informed the selection of proxy indicators, clarified the mathematical representation of variables, and improved the specification of inter-variable relationships. These refinements enhanced the model's credibility, relevance, and legitimacy.

In the next phase, study findings will be shared with farmers and local officials for validation and to explore policy implications, particularly around abandoned land revocation. As a living model, this approach demonstrates the value of participatory tools in evidence-based decision-making within dynamic food systems.

Keywords: peri-urban agriculture, land use